

Velocity Features as the Structural Invariant of Card-Testing Attacks in Online Payment Systems

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Abstract

Anti-fraud systems of online payment services are built on feature spaces assembled empirically — dozens to hundreds of attributes without a functional typology. We show that such spaces are structurally non-uniform and that a large part of their features carries no independent signal at all. The transaction feature space is organized into four functional families (temporal, velocity, amount-structure, identity) and their separability with respect to the fraud label is measured on 11,127 real card transactions by MANOVA, the silhouette index, sequential family ablation and stratified tests. The velocity family dominates (silhouette 0.81 against 0.06 for temporal features) and alone reproduces the quality of the full space (PR-AUC 0.551 against 0.547), while its removal costs 61 % of PR-AUC. Popular temporal features turn out to be indirect proxies of velocity behavior: the significant “weekend” effect (χ^2 11.73, $p < 0.001$) vanishes under stratification to the high-velocity subsample (χ^2 0.56, $p = 0.46$). The identified composite signature “at least two card transactions per hour with an amount repeated within a 10-minute window” holds an 80.6 % fraud rate at a 5.14 % base rate and covers 32.7 % of all fraud in 2.1 % of the volume, giving a compact detection rule for card-testing attacks. External validation on IEEE-CIS (590,540 transactions) confirms the structural principle but shows that the dominant behavioral family is regime-specific.

Keywords

card-testing attacks, payment fraud detection, behavioral patterns, feature engineering, social engineering

1. Introduction

Payment fraud in online services is a mass-scale attack surface rather than a set of isolated incidents. In 2023 Brazil alone recorded about 3.7 million fraudulent card transactions with aggregate losses of about 3.5 billion BRL, while the National Bank of Ukraine reports an annual growth of card-fraud losses. A substantial share of these losses begins with *card-testing*: an automated verification of stolen card credentials through a series of small transactions, which precedes the monetization of the compromised data and is documented in industry reports of Visa, Stripe and Adyen as one of the most widespread attacks against payment gateways. The credentials it verifies come predominantly from phishing and other social-engineering campaigns or from data breaches, so detecting the testing stage cuts the monetization path of those attacks.

Detection of such attacks is a pattern-recognition task in a high-dimensional feature space. In practice these spaces are built empirically: dozens to hundreds of attributes — time of day, day of week, amount, merchant, country, device — without a systematic typology. This overloads the model and invites interpretation errors, when a significant signal of one feature is in fact an indirect manifestation of a deeper behavioral variable. The industry belief about fraudster activity “at night” or “on weekends” is the typical case: such time slots merely coincide with periods of increased velocity activity of a compromised card.

The problem addressed here is therefore not the accuracy of another classifier but the *structure* of that feature space: which family carries the class separation, and which families only reproduce it indirectly.

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2. Novelty with Respect to Known Works

Classical work on card fraud treats feature engineering as auxiliary. Dal Pozzolo et al. [1] calibrate probabilities under severe class imbalance (fraud rate $< 1\%$); Jurgovsky et al. [2] apply LSTM to card transaction sequences; Lucas et al. [3] develop multi-perspective hidden Markov models that implicitly encode short-term patterns. Bahnsen et al. [4] give the first systematic study of feature-engineering strategies, aggregating attributes over time windows for a 1.5–3-fold quality gain; in APATE [5] card–merchant network features capture the graph structure of fraud propagation. On Brazilian data, Carneiro et al. [6] note repeated amounts as an artifact of stolen-card testing, while De Sá et al. [7] and Carcillo et al. [8] tune classifiers to the market's loss function and to streaming operation.

In all of these works features are *added*; which of them are structurally redundant is not asked. Card-testing itself, though well documented by payment systems, is rarely a separate analytical category in the literature: its markers (inter-transaction gaps, amount repetition) are detected among dozens of others, without emphasis on their invariance.

The novelty of this work is threefold. First, the transaction feature space is organized into functional families *a priori*, by the nature of the information a feature reflects, and separation strength is measured per family, not per feature. Second, the redundancy of popular temporal features is established not by importance ranking but by a stratified test that removes the velocity effect and shows the temporal signal to be a hidden common effect. Third, a single composite feature operationalizes the card-testing signature at an 80.6 % fraud rate — a rule usable alongside a model. The same family-wise methodology was previously applied by the authors outside the financial domain, to role-based patterns of Counter-Strike 2 players [9].

3. Feature Space and Methods

The public Predict Chargeback Frauds dataset [10] is used: 11,127 real transactions of 1–30 May 2015 with a masked card number, a second-precision timestamp, an amount and a chargeback label; 9,260 unique cards, fraud rate 5.14 % ($N = 572$). The transaction-to-card ratio of 1.20 indicates predominantly single-use cards, the many-transaction ones being tied to fraudulent activity.

Twenty-six features are grouped into four families by the nature of the information they reflect: **temporal** (7) — cyclically encoded hour and day of week, *weekend*, *night*, *afternoon_peak*; **velocity** (10) — the gap to the card's previous transaction, cumulative and rolling counts and sums in 1-, 6- and 24-hour windows, repetition of the previous amount and of an amount within a 10-minute window; **amount-structure** (7) — logarithm, roundness modulo 10/50/100, fractional part, thresholds; **identity** (2) — Bayesian-smoothed target encodings of the BIN and the amount. Velocity features respect causality: for the i -th transaction of a card they use only prior transactions of the same card, and identity encodings are fitted on the training part only.

The space was analyzed by PCA; univariate ANOVA and the Mann–Whitney U test; MANOVA (Wilks' λ) per family; the silhouette index [12] for geometric separation; logistic regression and gradient boosting on a time-ordered 75/25 split with sequential family ablation and bootstrap intervals for PR-AUC (1000 resamples); permutation importance; and a χ^2 stratification test for hidden common effects.

4. Results

Structural non-uniformity of the space. All four families are statistically significant by MANOVA ($p < 10^{-30}$), which at $N = 11,127$ is uninformative on its own, so the conclusions rest on effect size. By that criterion the families differ by an order of magnitude: velocity (Wilks' $\lambda = 0.682$) and identity (0.659) against temporal (0.986) and amount (0.975), where λ close to 1 means that within-group variation overrides the between-group one. The silhouette index gives the same picture in a sharper form: velocity +0.81, identity +0.65, amount +0.22, temporal +0.06 (Fig. 1). The

index of the identity family is inflated, since target-encoded features carry direct information about the label; the honest measure of natural behavioral separability is therefore the velocity index. The space is not one-dimensional: the first two principal components explain only 37.2 % of the variance, so the behavior is not reducible to a single “fraud score”.

Ablation. Sequential removal of families (Table 1) confirms the hierarchy. Removing velocity is catastrophic: PR-AUC falls from 0.547 to 0.212 (−61 %), while removing any other family costs at most 6 %. The velocity family alone reproduces the full space (PR-AUC 0.551 against 0.547, precision in the top 1 % of risk 0.926, recall in the top 5 % even higher, 0.511 against 0.454) — the remaining 16 features add no independent signal. Temporal features alone are close to a random classifier (PR-AUC 0.073 at a base rate of 0.062). Bootstrap 95 % intervals separate without-velocity 0.212 [0.165; 0.276] from full space 0.547 [0.478; 0.616] and velocity only 0.551 [0.474; 0.630], which almost fully overlap. Gradient boosting is qualitatively identical (0.422 full, 0.178 without velocity, 0.535 velocity only). Permutation importance is consistent: *bin_te* (identity) ranks first at Δ PR-AUC 0.256 but is inflated by target-encoding memorization, and the strongest purely behavioral features are *gap_min_prev* (velocity) 0.190 and *n_tx_card_1h* 0.039, against at most 0.006 for every temporal feature.

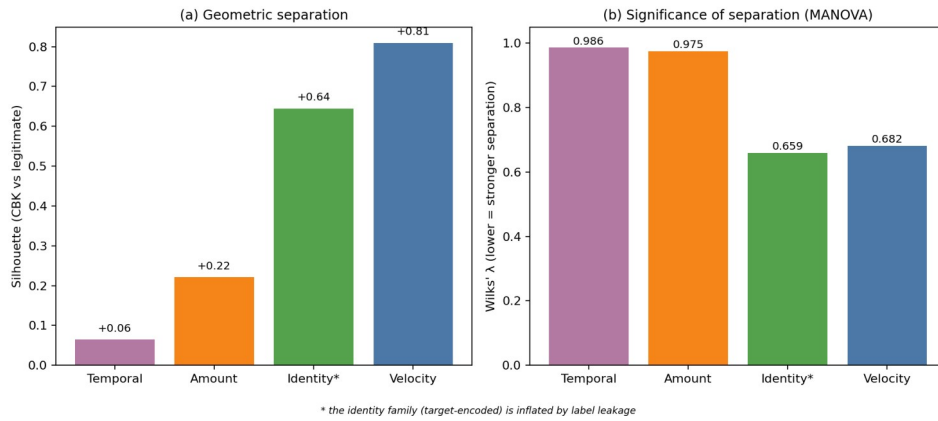


Figure 1: Discriminative power by feature families: (a) silhouette index, higher is better; (b) Wilks' λ from MANOVA, lower means stronger separation. The velocity family dominates; the identity family (marked with an asterisk) is inflated by label leakage through target encoding.

Table 1

Sequential feature-family ablation (logistic regression, time-ordered 75/25 split). Precision and recall are measured on the top 1 % and top 5 % highest-risk transactions.

Configuration	ROC-AUC	PR-AUC	Prec., top 1 %	Recall, top 5 %
Full space (26 features)	0.875	0.547	0.926	0.454
Without temporal (19)	0.877	0.531	0.926	0.443
Without velocity (16)	0.738	0.212	0.333	0.241
Without amount (19)	0.874	0.515	0.889	0.431
Temporal only (7)	0.510	0.073	0.000	0.029
Velocity only (10)	0.845	0.551	0.926	0.511
Amount only (7)	0.602	0.102	0.000	0.046

Temporal features as a hidden common effect. On the full dataset the “weekend” effect is statistically significant: $\chi^2 = 11.73$, $p = 6.1 \cdot 10^{-4}$, with a fraud rate of 6.74 % on weekends against 4.81 % on weekdays (1.31× over the base). Under stratification to the high-velocity subsample ($n_tx_card_24h \geq 2$), however, it vanishes: $\chi^2 = 0.56$, $p = 0.456$, and the rates within that subsample become indistinguishable (67.0 % against 71.9 %). The day of the week correlates with fraud only because high-velocity bursts occur more often on weekends, when monitoring is relaxed. Correlation analysis of the residuals of a model trained on the velocity family alone agrees: all temporal features have a partial $|r| < 0.1$ except *weekend* ($r = -0.24$). The velocity family thus absorbs about 95 % of the temporal signal.

The card-testing signature. The composite feature “at least 2 card transactions per hour with the amount repeated within a 10-minute window” ($n_tx_card_1h \geq 2$ AND $same_amount_10min = 1$) covers 232 transactions, i.e. 2.1 % of the volume, with a fraud rate of 80.6 % among them (187 of 232) — a 15.7-fold lift over the base rate — and captures 32.7 % of all fraudulent transactions of the dataset. It is invariant in a concrete empirical sense: it depends neither on the day of the week ($\chi^2 = 0.56$, $p = 0.46$ on the high-velocity subsample), nor on the time of day, nor on the specific amount, since only the repetition matters, not the value. Two percent of the traffic thus contains a third of all fraud, which makes the rule directly deployable as a real-time complement to a scoring model.

External validation. On the independent IEEE-CIS Fraud Detection dataset [11] (590,540 e-commerce transactions, fraud rate 3.50 %) a direct transfer degenerates: the field *card1* is a coarse proxy aggregating on average 43.6 transactions, so per-card velocity features lose their behavioral meaning (silhouette 0.03) — an artifact of the entity mapping rather than a disappearance of the signal. Studied on the dataset’s own engineered features, the structural principle holds: the relationship counters *C1–C14* give the strongest geometric separation (silhouette 0.48) and the time deltas *D1–D15* the highest mean F-statistic (1530), while amount-structure (0.07) and temporal (0.03) features remain weak, exactly as in the main dataset. The fraud regime, however, differs: the rate rises for new cards ($D1 \leq 7$ days — 9.4 % against 1.2 % for cards older than a year) and for entities with many relations ($C2 > 10$ — 9.65 % against 2.3 %), which is an account/identity signature rather than velocity card-testing. Invariant is therefore the structural principle, not a particular family.

5. Conclusions

A taxonomy of four functional families of behavioral features of card transactions was proposed and their separability measured on 11,127 real transactions. Fraudulent transactions form a statistically significantly separated region of the space (Wilks’ $\lambda = 0.547$, $F = 352.95$, $p < 10^{-300}$) that is not reducible to a one-dimensional score.

The space is structurally non-uniform: velocity features (silhouette 0.81) dominate over identity (0.65), amount-structure (0.22) and temporal ones (0.06); removing the velocity family costs 61 % of PR-AUC, while that family alone reproduces the quality of the full space. Popular temporal features are indirect proxies of the velocity pattern rather than independent signals — the significant “weekend” effect disappears under stratification to the high-velocity subsample, and alone such features classify at the level of chance.

The composite card-testing signature (at least two transactions of a card per hour with an amount repeated within a 10-minute window) holds an 80.6 % fraud rate and covers 32.7 % of all fraud within 2.1 % of the volume. Practically, the structural core of an anti-fraud feature space should be built from velocity features, temporal and formal amount features being auxiliary, and the identified signature can serve as a cheap, highly precise rule-based detector of card-testing next to a model. External validation on IEEE-CIS confirms the principle of non-uniform distribution of separation strength across behavioral families but shows the dominant mechanism to be regime-specific: velocity repetition for single-use cards against entity relatedness and card age in card-not-present e-commerce. **Limitations.** The main dataset covers a single month of one payment

gateway and holds no merchant, device or geolocation fields, so the signature is established for the card-testing regime only; the identity family is measured through target encodings, whose power is partly memorization, which is why the conclusions rest on the velocity family; and IEEE-CIS has no true card identifier. Further work targets a taxonomy invariant to the fraud regime, validation on data with a true card identifier, an evaluation of the signature under adversarial adaptation of the attacker's pace, and the dynamics of the invariant features in a streaming setting.

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